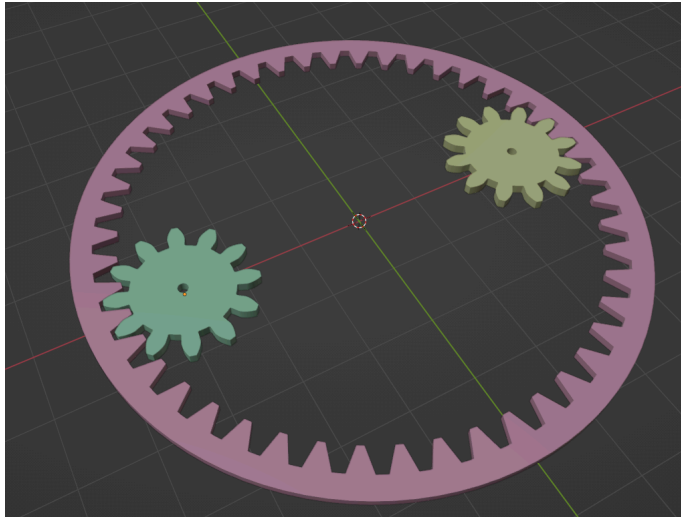


For our horizontal mechanism, our group planned to use a ring gear (until iteration 2, when it was changed), not too dissimilar to the elevation mechanism. As such, the principles behind both horizontal and elevation mechanisms are explained here.

Gears are typically used to transmit rotation of one axis to the axis of another. The teeth of the gear mesh together, so that once a gear rotates, the adjacent gear's teeth are pushed by the teeth of the original gear. As a result, this applies a torque on the adjacent gear, causing it to spin. Because of Newton's third law, however, the original gear also has a torque applied on it, thus spins slower under the same force, compared to if it was not powering another gear. Because of this equal torque, however, the rotational speed of each gear is constant where they mesh together. Because $\tau = rF$, the larger a gear, the slower it will spin, but the more force it will have while it spins. $\tau = \tau \therefore r_1 F_1 = r_2 F_2 \therefore F_2 = \frac{r_1}{r_2} F_1$ This allows for gear-ups and gear-downs, allowing for greater force applied with little exertion, although more distance has to be traveled to perform the same movement. As radius is analogous to the circumference (and the number of teeth along the surface of the gear), this ratio of forces can be simplified to gear ratios, determined by the number of teeth of each gear, assuming the teeth are the same shape, and will mesh properly.

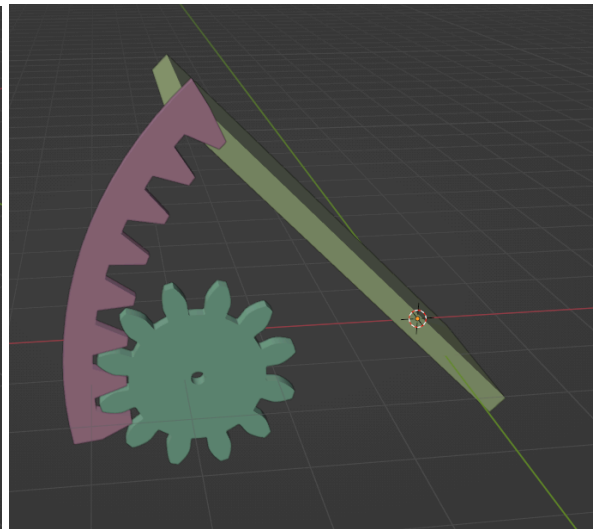
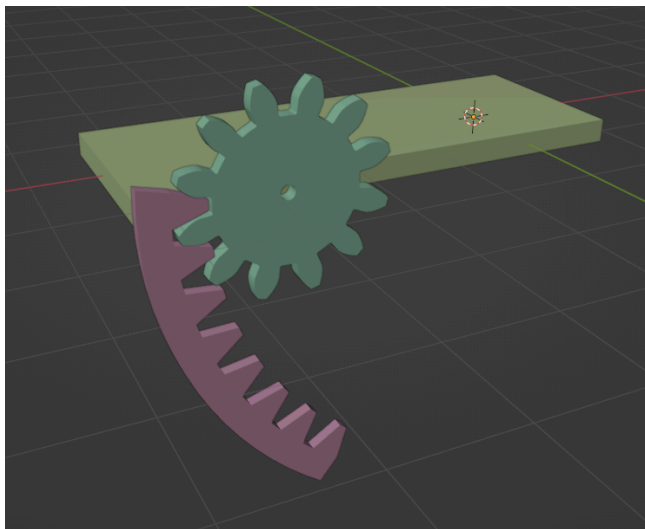
This concept also applies to ring gears, except the rotation of the original gear exerts a force on the ring gear surrounding it. Once again, the torques the ring gear and spur gear are able to exert depend on the number of teeth, but in our case, we planned to use a ring gear to transfer the motion of a gear. This was done such that we wouldn't need to use a belt or chain of gears to move the rotational power of a motor to a location where placing a motor would be unfeasible. As such, we planned to use this to allow the base to be rotated from underneath, such that we

wouldn't have to stack mechanisms atop one another (see image below).

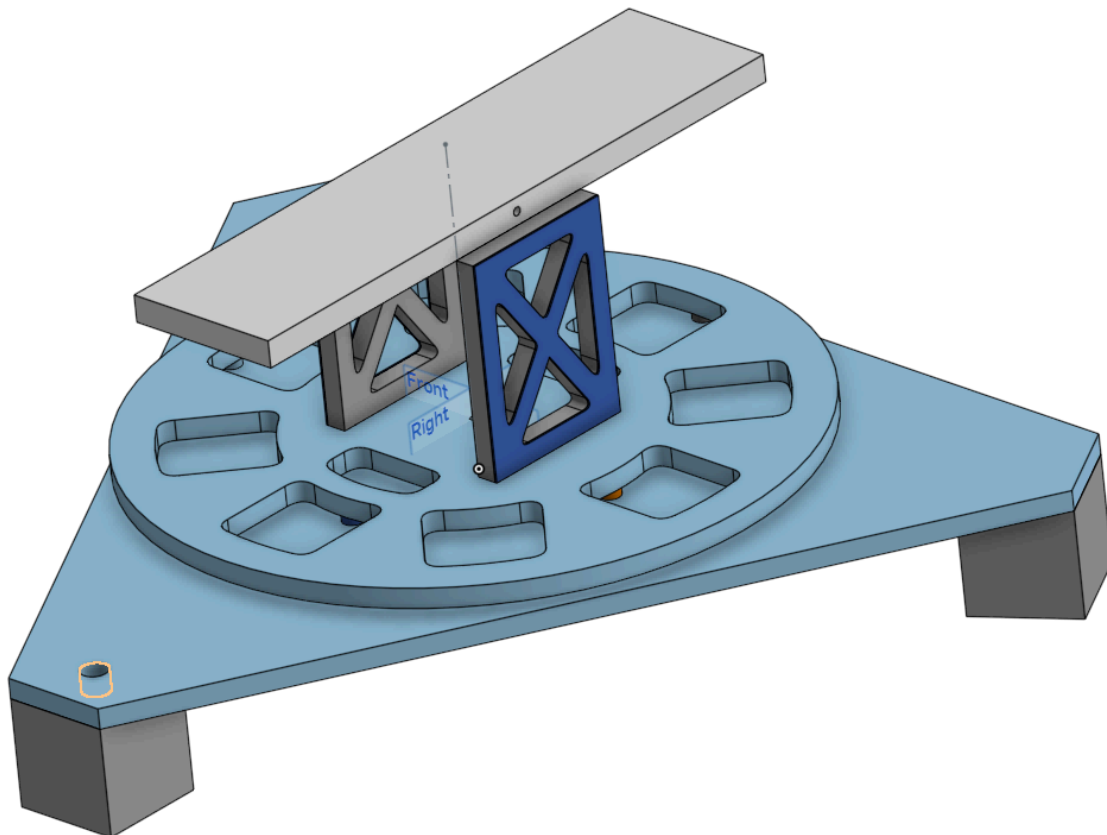


As an example, this was the model for the small scale prototype (the STL files are brought into blender for the sake of visibility of the individual meshes). The yellow gear would spin at the same rate as the green gear, and the green gear would spin at the same rate as the yellow gear.

The principle for the vertical elevation mechanism follows the ideas used with the ring gear; however, this would only use a small “portion” of the ring gear.



This was a modified version of the ring gear modified to be used with the elevation mechanism. It would be attached such that once the green gear was rotated, it would raise the purple ring gear; as a result, In actual usage, the purple ring gear's section would rest on the green gear, acting as a constant force requiring for the motor to constantly be active. To avoid this, we planned to have the base elevate itself only before firing, and have it return to its resting position immediately after. Below is an image of the mockup in onshape. Asking Bill Rodriguez, he showed approval for the basic ideas at hand, but concerned the launcher (that was to be placed atop the launcher) would have to be balanced precisely to reduce stress on the elevation system. While this wasn't stressed early on, this was eventually accounted for in prototype 3, before the elevation system was outright expunged from prototype 4 (for both complexity and time).



The entire rotating portion of the system was placed atop a 6" lazy susan, with nylon spacers in between the rotating portion and the stationary base. The nylon spacers were used so less of the gravitational force would be placed atop the lazy susan, and the spacers were chosen to be nylon such that friction would also be reduced. The frames and rotary base were planned to be made of a 0.75" MDF board to improve the structural integrity of the base. Since the weight was worryingly unconfirmed by the Andean Bear Launcher team, extra 1/16" aluminum frames were also added to the sides of the vertical frames.